HOW FEW DIRECTIONS DOES AN OPERATOR NEED? ORBIT FILTERS, SYMBOL STRUCTURE, AND SHARED SCHEDULES

THEORY NOTE ACCOMPANYING

RANK-OPTIMAL DIRECTIONAL TAYLOR JETS FOR HIGH-ORDER MIXED DERIVATIVES

Abstract. The directional schedule of the main paper is built from roots of unity, and it is minimal for a single monomial derivative. This note asks what that construction is really doing and how far it goes. Three answers follow. First, the roots-of-unity grid is not specific to monomials at all: for an arbitrary homogeneous constant-coefficient operator it produces an exact schedule whose coefficients are a discrete Fourier transform of the symbol, with no linear solve and no dimension restriction. Second, that grid is a complete intersection, and complete intersections are provably far from minimal once three or more coordinates are active; for a generic symbol the grid costs about $N!$ times the minimum in N active variables. Third, both the roots-of-unity schedule and its correct generalization are instances of a single principle: a character-weighted orbit of a finite group reproduces any symbol lying in a one-dimensional isotypic component. Choosing an abelian group recovers the Fourier grid; choosing a nonabelian group turns the design problem for powers of the Laplacian into the classical problem of cubature on the sphere, whose solutions are real, positively weighted, and in the smallest cases provably minimal. We also record that scheduling several operators, or a whole derivative tensor, on one shared direction set is exactly optimal at length equal to the dimension of the span, which quantifies how much term-by-term scheduling wastes.

1. Scope and notation. Throughout, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and $\boldsymbol{z} = (z_0, \dots, z_n)$ are coordinates on the active subspace of the problem, so that $N := n + 1$ is the number of active variables. We write $\mathbb{K}[\boldsymbol{z}]_p$ for the homogeneous polynomials of degree p, whose dimension is

$$D_{p,N} \;:=\; \dim \mathbb{K}[\boldsymbol{z}]_p \;=\; \binom{p+N-1}{N-1}. \tag{1.1}$$

A homogeneous constant-coefficient operator of order p and its symbol are

$$\mathcal{P} \;=\; \sum_{|\beta|=p} b_\beta\, \partial^\beta, \qquad \sigma(\boldsymbol{z}) \;=\; \sum_{|\beta|=p} b_\beta\, \boldsymbol{z}^\beta \in \mathbb{K}[\boldsymbol{z}]_p. \tag{1.2}$$

For a direction $\boldsymbol{v}$ we write $\boldsymbol{v} \cdot \boldsymbol{z} = \sum_j v_j z_j$, and $T_p(\boldsymbol{x}; \boldsymbol{v})$ for the order-p coefficient of $\tau \mapsto u(\boldsymbol{x} + \tau \boldsymbol{v})$, as in the main paper. A *directional schedule* for $\mathcal{P}$ of length R is a family $(c_r, \boldsymbol{v}_r)_{r=1}^R$ with $\mathcal{P}u(\boldsymbol{x}) = \sum_r c_r\, T_p(\boldsymbol{x}; \boldsymbol{v}_r)$ for all admissible u. We take the following bridge as given; it is Proposition 3.6 of the main paper.

Proposition 1.1 (Bridge). *A family $(c_r, \boldsymbol{v}_r)_{r=1}^R$ is a directional schedule for $\mathcal{P}$ if and only if*

$$p!\, \sigma(\boldsymbol{z}) \;=\; \sum_{r=1}^{R} c_r\, (\boldsymbol{v}_r \cdot \boldsymbol{z})^p \qquad \text{in } \mathbb{K}[\boldsymbol{z}]_p, \tag{1.3}$$

and the least length of a schedule over $\mathbb{K}$ is the Waring rank $R_{\mathbb{K}}(\sigma)$.

Everything below is therefore a statement about writing one fixed form σ as a short combination of pth powers of linear forms. Two standard devices are used. Let $\mathbb{K}[\boldsymbol{y}] = \mathbb{K}[y_0, \dots, y_n]$ act on $\mathbb{K}[\boldsymbol{z}]$ by $y_j \mapsto \partial/\partial z_j$. The *apolar ideal* of σ is

$$\sigma^\perp \;:=\; \{\Theta \in \mathbb{K}[\boldsymbol{y}] \;:\; \Theta(\partial)\, \sigma = 0\}, \tag{1.4}$$

a homogeneous ideal, and the *catalecticant* in degree q is the linear map

$$\mathrm{Cat}_q(\sigma)\colon \mathbb{K}[\boldsymbol{y}]_q \to \mathbb{K}[\boldsymbol{z}]_{p-q}, \qquad \Theta \mapsto \Theta(\partial)\sigma, \qquad \ker \mathrm{Cat}_q(\sigma) = (\sigma^\perp)_q. \tag{1.5}$$

We use the apolarity lemma in the form: for a finite reduced $X \subset \mathbb{P}(\mathbb{K}^N)$ with homogeneous vanishing ideal I_X,

$$\sigma \in \mathrm{span}\{(\boldsymbol{v}\cdot\boldsymbol{z})^p : [\boldsymbol{v}] \in X\} \iff I_X \subseteq \sigma^\perp. \tag{1.6}$$

Finally, $U_m := \{\zeta \in \mathbb{C} : \zeta^m = 1\}$, and we use repeatedly that

$$\sum_{\zeta \in U_m} \zeta^k \;=\; m \cdot \mathbf{1}_{m|k}, \qquad k \in \mathbb{Z}. \tag{1.7}$$

2. The Fourier grid is not about monomials. The schedule of the main paper attaches roots of unity to a monomial $\boldsymbol{z}^a$. The next theorem shows that the same point set serves an arbitrary symbol, with coefficients given in closed form by a discrete Fourier transform. No linear system is solved, and no restriction is placed on N.

THEOREM 2.1 (Universal Fourier schedule). *Let $\sigma \in \mathbb{K}[\boldsymbol{z}]_p$ be nonzero with coefficients b_β as in* (1.2). *Put*

$$d_j := \deg_{z_j} \sigma, \qquad m_j := d_j + 1, \qquad j = 0, \dots, n, \tag{2.1}$$

and relabel the coordinates so that $m_0 = \min_j m_j$. Set $M := \prod_{j=1}^n m_j$. For $\zeta = (\zeta_1, \dots, \zeta_n) \in U_{m_1} \times \cdots \times U_{m_n}$ define

$$\boldsymbol{v}_\zeta \;:=\; e_0 + \sum_{j=1}^n \zeta_j\, e_j, \qquad c_\zeta \;:=\; \frac{1}{M} \sum_{|\gamma|=p} \gamma!\, b_\gamma \prod_{j=1}^n \zeta_j^{-\gamma_j}. \tag{2.2}$$

Then $(c_\zeta, \boldsymbol{v}_\zeta)_\zeta$ is a directional schedule for $\mathcal{P}$ of length M.

Proof. By Proposition 1.1 it suffices to verify (1.3). Since $\boldsymbol{v}_\zeta \cdot \boldsymbol{z} = z_0 + \sum_{j \ge 1} \zeta_j z_j$, the multinomial theorem gives

$$\sum_\zeta c_\zeta (\boldsymbol{v}_\zeta \cdot \boldsymbol{z})^p = \sum_{|\beta|=p} \frac{p!}{\beta!}\, \boldsymbol{z}^\beta \underbrace{\sum_\zeta c_\zeta \prod_{j=1}^n \zeta_j^{\beta_j}}_{=:\, S_\beta},$$

so (1.3) is equivalent to $S_\beta = \beta!\, b_\beta$ for every β with $|\beta| = p$. Substituting (2.2) and applying (1.7) in each factor,

$$S_\beta = \frac{1}{M} \sum_{|\gamma|=p} \gamma!\, b_\gamma \prod_{j=1}^n \Big(\sum_{\zeta_j \in U_{m_j}} \zeta_j^{\beta_j - \gamma_j} \Big) = \sum_{|\gamma|=p} \gamma!\, b_\gamma \prod_{j=1}^n \mathbf{1}_{m_j \,|\, \beta_j - \gamma_j}. \tag{2.3}$$

We claim that for γ with $b_\gamma \neq 0$ the product of indicators in (2.3) equals 1 precisely when $\gamma = \beta$. Sufficiency is clear. For necessity, write $\beta_j - \gamma_j = k_j m_j$ with $k_j \in \mathbb{Z}$ for $j = 1, \dots, n$. Because $b_\gamma \neq 0$, the definition (2.1) forces $\gamma_j \le d_j = m_j - 1$ for *every* j, including $j = 0$. If $k_j \le -1$ for some $j \ge 1$, then $\gamma_j \ge \beta_j + m_j \ge m_j > d_j$, a contradiction; hence $k_j \ge 0$ for all $j \ge 1$. Summing $|\beta| = |\gamma| = p$ gives $\sum_{j \ge 0} (\beta_j - \gamma_j) = 0$, that is,

$$\sum_{j=1}^n k_j m_j \;=\; \gamma_0 - \beta_0 \;\le\; \gamma_0 \;\le\; d_0 \;=\; m_0 - 1. \tag{2.4}$$

If some $k_j \geq 1$ then the left side of (2.4) is at least $m_j \geq m_0$, contradicting the bound $m_0 - 1$. Therefore $k_j = 0$ for all $j \geq 1$, so $\beta_j = \gamma_j$ for $j \geq 1$, and then $\beta_0 = \gamma_0$ because $|\beta| = |\gamma|$. This proves the claim.

Consequently the sum (2.3) retains at most the single term $\gamma = \beta$. If $b_\beta \neq 0$ it retains exactly that term and $S_\beta = \beta!\, b_\beta$. If $b_\beta = 0$, no γ with $b_\gamma \neq 0$ survives, since surviving would force $\gamma = \beta$ and hence $b_\beta \neq 0$; thus $S_\beta = 0 = \beta!\, b_\beta$. ◻

Two features of Theorem 2.1 deserve comment.

Remark 2.2 (The base coordinate must be one of least degree). *The relabelling $m_0 = \min_j m_j$ is not cosmetic; it is exactly what (2.4) consumes. Take $\sigma = z_0^4 + z_0 z_1^3$, so $p = 4$, $d_0 = 4$, $d_1 = 3$ and the least degree occurs at $j = 1$. Building the grid on the base $j = 0$ instead uses $\zeta \in U_4$ and the pair $\beta = (0, 4)$, $\gamma = (4, 0)$ satisfies $4 \mid \beta_1 - \gamma_1$ with $\gamma \neq \beta$. The aliased term survives and the construction returns $\sigma + z_1^4$ rather than σ. Choosing the base correctly makes every alias impossible at once.*

Remark 2.3 (Recovering the monomial case). *For $\sigma = \boldsymbol{z}^a$ with a the active exponent vector, only $\gamma = a$ contributes to (2.2), so $c_\zeta = (a!/M) \prod_j \zeta_j^{-a_j}$ and $M = \prod_{j \geq 1}(a_j + 1)$. This is Theorem 3.4 of the main paper, up to replacing ζ_j by ζ_j^{-1}, which permutes the grid. So the monomial schedule is the one-term instance of Theorem 2.1.*

The practical reading of Theorem 2.1 is that the grid is a *universal* device: it needs nothing from σ beyond its coordinate degrees, it never solves a linear system, and its length $M = \prod_{j \geq 1} m_j$ is available in advance. The rest of this note is about the price of that universality.

3. What grid structure costs. The grid of Theorem 2.1 is the zero set of the n binomials $y_j^{m_j} - y_0^{m_j}$, $j = 1, \ldots, n$, hence a complete intersection in $\mathbb{P}^n$. That single structural fact bounds how good it can ever be.

Definition 3.1 (Initial degree). *For $0 \neq \sigma \in \mathbb{K}[\boldsymbol{z}]_p$ set $\delta(\sigma) := \min\{q \geq 1 : (\sigma^\perp)_q \neq 0\}$.*

Proposition 3.2 (Complete intersections are large). *Let $X \subset \mathbb{P}^n$ be a finite apolar scheme for σ, that is $I_X \subseteq \sigma^\perp$, and suppose X is a complete intersection of n forms of degrees $e_1, \ldots, e_n$. Then*

$$|X| \;=\; \prod_{i=1}^{n} e_i \;\geq\; \delta(\sigma)^n. \tag{3.1}$$

Proof. The n defining forms lie in $I_X \subseteq \sigma^\perp$, so each has degree at least $\delta(\sigma)$ by Definition 3.1. Since X is a complete intersection of these forms, Bézout's theorem gives $\deg X = \prod_i e_i$, and X is finite and reduced so $|X| = \deg X$. ◻

It remains to locate $\delta(\sigma)$ for a symbol with no special structure.

Lemma 3.3 (Initial degree of a generic symbol). *For σ generic in $\mathbb{K}[\boldsymbol{z}]_p$ one has $\delta(\sigma) = \lfloor p/2 \rfloor + 1$.*

Proof. By (1.5), $(\sigma^\perp)_q \neq 0$ if and only if $\mathrm{Cat}_q(\sigma)$ has nontrivial kernel, so the claim is that a generic σ has Cat_q of maximal rank $\min(D_{q,N}, D_{p-q,N})$ for every q. Rank is lower semicontinuous, and there are finitely many values of q, so it suffices to exhibit for each q one form of maximal rank.

We use one fact throughout: any $k \leq D_{d,N}$ generic points of the degree-d Veronese are linearly independent. This holds because the Veronese spans $\mathbb{P}(\mathbb{K}[\boldsymbol{z}]_d)$, so some $D_{d,N}$ of its points are independent, and independence is an open condition. Applied to $\boldsymbol{z}$ it says that generic powers $\ell_1^d, \ldots, \ell_k^d$ are independent; applied to the dual variables it says that k generic points $\ell_1, \ldots, \ell_k$ impose independent conditions on $\mathbb{K}[\boldsymbol{y}]_d$, so that $\Theta \mapsto (\Theta(\ell_i))_i$ is onto $\mathbb{K}^k$.

Fix q and set $s := \min(D_{q,N}, D_{p-q,N})$; take $\sigma := \sum_{i=1}^{s} \ell_i^p$ with generic ℓ_i. Since $\Theta(\partial)\ell^p = \frac{p!}{(p-q)!}\Theta(\ell)\,\ell^{p-q}$ for $\Theta \in \mathbb{K}[\boldsymbol{y}]_q$,

$$\mathrm{Cat}_q(\sigma)(\Theta) \;=\; \frac{p!}{(p-q)!}\sum_{i=1}^{s}\Theta(\ell_i)\,\ell_i^{p-q}. \tag{3.2}$$

If $q \leq p-q$ then $s = D_{q,N} \leq D_{p-q,N}$, so the powers ℓ_i^{p-q} are independent; a kernel element would force $\Theta(\ell_i) = 0$ for every i, and the $s = D_{q,N}$ generic points impose independent conditions in degree q, whence $\Theta = 0$. Thus Cat_q is injective of rank $D_{q,N}$, which is maximal. If $q > p-q$ then $s = D_{p-q,N} \leq D_{q,N}$, the vector $(\Theta(\ell_i))_i$ ranges over all of $\mathbb{K}^s$, and the s independent powers ℓ_i^{p-q} form a basis of $\mathbb{K}[\boldsymbol{z}]_{p-q}$; by (3.2) the map is onto, of maximal rank $D_{p-q,N}$.

A maximal-rank Cat_q has nonzero kernel exactly when $D_{q,N} > D_{p-q,N}$, that is when $q > p-q$, and the least such q is $\lfloor p/2 \rfloor + 1$. ∎

Corollary 3.4 (The price of grid structure). *Fix $N \geq 2$ and let $\sigma \in \mathbb{K}[\boldsymbol{z}]_p$ be generic. Every complete intersection apolar scheme for σ has at least $(\lfloor p/2 \rfloor + 1)^{N-1}$ points, while $R_{\mathbb{C}}(\sigma)$ is asymptotic to $D_{p,N}/N$ as $p \to \infty$. Consequently*

$$\lim_{p\to\infty}\frac{(\lfloor p/2\rfloor+1)^{N-1}}{R_{\mathbb{C}}(\sigma)} \;=\; \frac{N!}{2^{\,N-1}}, \qquad \textit{and} \qquad \lim_{p\to\infty}\frac{M}{R_{\mathbb{C}}(\sigma)} \;=\; N!, \tag{3.3}$$

where $M = (p+1)^{N-1}$ is the length of the grid of Theorem 2.1 for a generic symbol.

Proof. The first bound is Proposition 3.2 together with Lemma 3.3. The generic Waring rank is $\lceil D_{p,N}/N \rceil$ outside the finite list of Alexander–Hirschowitz exceptions, and $D_{p,N} = \binom{p+N-1}{N-1} \sim p^{N-1}/(N-1)!$, so $R_{\mathbb{C}}(\sigma) \sim p^{N-1}/N!$. Dividing $(\lfloor p/2 \rfloor + 1)^{N-1} \sim (p/2)^{N-1}$ by this gives $N!/2^{N-1}$. A generic σ has $d_j = p$ for every j, so $M = (p+1)^{N-1} \sim p^{N-1}$, and dividing gives $N!$. ∎

The two limits in (3.3) separate two different losses. The factor $N!/2^{N-1}$ is the intrinsic cost of insisting on complete intersection structure, and the remaining factor 2^{N-1} is the cost of building the grid from the coordinate degrees rather than from $\sigma^{\perp}$. Both vanish when $N = 2$: there $N!/2^{N-1} = 1$, which is the algebraic reason binary forms are easy, since every finite subscheme of $\mathbb{P}^1$ is cut out by a single form and the complete intersection hypothesis is vacuous. For $N = 3$ the intrinsic factor is $3/2$, for $N = 4$ it is 3, for $N = 5$ it is $15/2$, and it grows superexponentially.

Remark 3.5 (When the grid is exactly minimal). *Suppose $\sigma = \ell_0^{a_0} \cdots \ell_n^{a_n}$ for linearly independent linear forms ℓ_j, that is, σ is projectively equivalent to a monomial. Waring rank is invariant under invertible linear changes of variables, so running Theorem 2.1 in the coordinate system dual to (ℓ_j) yields a schedule of length $\prod_{j\geq 1}(a_j + 1)$, which is the rank of that monomial and hence minimal. Corollary 3.7 of the main paper is the case $\sigma = (z_1^2 + z_2^2)^m = (z_1 + \mathrm{i}z_2)^m (z_1 - \mathrm{i}z_2)^m$ in two variables. The hypothesis is restrictive: forms equivalent to monomials constitute a subvariety of dimension at most N^2 inside the $D_{p,N}$-dimensional space of all symbols.*

4. Orbit filters. Identity (1.7) is character orthogonality for the cyclic group U_m. Read that way, the roots-of-unity construction is one instance of a principle that does not require the group to be abelian, and the nonabelian instances are the interesting ones. Throughout this section G is a finite subgroup of $GL_N(\mathbb{K})$ acting on $\mathbb{K}[\boldsymbol{z}]_p$ by

$$(g \cdot F)(\boldsymbol{z}) \;:=\; F(g^{\mathsf{T}}\boldsymbol{z}). \tag{4.1}$$

This is a left action, since $(gh)^{\mathsf{T}} = h^{\mathsf{T}} g^{\mathsf{T}}$, and it is the action matched to the pairing: because $\boldsymbol{v} \cdot g^{\mathsf{T}}\boldsymbol{z} = (g\boldsymbol{v}) \cdot \boldsymbol{z}$, it satisfies

$$g \cdot (\boldsymbol{v} \cdot \boldsymbol{z})^p \;=\; \big((g\boldsymbol{v}) \cdot \boldsymbol{z}\big)^p \tag{4.2}$$

for every g, with no orthogonality assumed. When G happens to be orthogonal, $g^{\mathsf{T}} = g^{-1}$ and (4.1) is the familiar substitution $F(g^{-1}\boldsymbol{z})$. We write $\pi_G := |G|^{-1} \sum_{g \in G} g$ for the Reynolds operator, a projection of $\mathbb{K}[\boldsymbol{z}]_p$ onto the invariants $\mathbb{K}[\boldsymbol{z}]_p^G$.

Definition 4.1 (Orbit average). *For $\boldsymbol{v} \in \mathbb{K}^N$ and a linear character $\chi \colon G \to \mathbb{K}^\times$ put*

$$A_{\boldsymbol{v}}^{\chi}(\boldsymbol{z}) \;:=\; \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)}\, \big((g\boldsymbol{v}) \cdot \boldsymbol{z}\big)^p, \tag{4.3}$$

and write $A_{\boldsymbol{v}} := A_{\boldsymbol{v}}^{\mathbf{1}}$ for the trivial character. The isotypic component *of χ is $\mathbb{K}[\boldsymbol{z}]_p^{(\chi)} := \{F : g \cdot F = \chi(g) F$ for all $g\}$; for $\chi = \mathbf{1}$ it is $\mathbb{K}[\boldsymbol{z}]_p^G$.*

Lemma 4.2 (Orbit averages are isotypic). *$A_{\boldsymbol{v}}^{\chi} \in \mathbb{K}[\boldsymbol{z}]_p^{(\chi)}$ for every $\boldsymbol{v}$ and every linear character χ. Moreover $A_{\boldsymbol{v}}^{\chi}$ is the image of $(\boldsymbol{v} \cdot \boldsymbol{z})^p$ under the projection $\pi_G^{\chi} := |G|^{-1} \sum_g \overline{\chi(g)}\, g$ onto $\mathbb{K}[\boldsymbol{z}]_p^{(\chi)}$, and consequently $\{A_{\boldsymbol{v}}^{\chi} : \boldsymbol{v} \in \mathbb{K}^N\}$ spans $\mathbb{K}[\boldsymbol{z}]_p^{(\chi)}$.*

Proof. Applying h to (4.3) and using (4.2) twice,

$$h \cdot A_{\boldsymbol{v}}^{\chi} = \frac{1}{|G|} \sum_g \overline{\chi(g)}\; h \cdot \big((g\boldsymbol{v}) \cdot \boldsymbol{z}\big)^p = \frac{1}{|G|} \sum_g \overline{\chi(g)} \big((hg\boldsymbol{v}) \cdot \boldsymbol{z}\big)^p = \chi(h)\, A_{\boldsymbol{v}}^{\chi},$$

the last step substituting $g \mapsto h^{-1}g$ and using $\overline{\chi(h^{-1}g)} = \chi(h)\overline{\chi(g)}$, valid because a linear character of a finite group takes values in the roots of unity. The same use of (4.2) shows $\pi_G^{\chi}\big((\boldsymbol{v} \cdot \boldsymbol{z})^p\big) = A_{\boldsymbol{v}}^{\chi}$. Since powers of linear forms span $\mathbb{K}[\boldsymbol{z}]_p$ and π_G^{χ} is a surjection onto $\mathbb{K}[\boldsymbol{z}]_p^{(\chi)}$, the images $A_{\boldsymbol{v}}^{\chi}$ span that component. ∎

Theorem 4.3 (Orbit filter). *Let $\boldsymbol{v} \in \mathbb{K}^N$ with orbit $X := G\boldsymbol{v}$, and let χ be a linear character of G.*

- (i) *If σ is a nonzero scalar multiple of $A_{\boldsymbol{v}}^{\chi} \neq 0$, then $\mathcal{P}$ admits a directional schedule supported on X, whose coefficient at the direction $g\boldsymbol{v}$ is $\lambda \sum_{h \in \mathrm{Stab}(\boldsymbol{v})} \overline{\chi(gh)}$ for a constant λ. Its length is $|X| = |G| \,/\, |\mathrm{Stab}(\boldsymbol{v})|$, and $|X|/2$ if p is even and $-I \in G$.*
- (ii) *In particular, if $\dim_{\mathbb{K}} \mathbb{K}[\boldsymbol{z}]_p^{(\chi)} = 1$ then the hypothesis of* (i) *holds for every σ in that component and every $\boldsymbol{v}$ with $A_{\boldsymbol{v}}^{\chi} \neq 0$, with no equation to solve.*
- (iii) *If $\sigma = \sum_\chi \sigma^{(\chi)}$ is the decomposition of a general σ into components over the linear characters, and each $\sigma^{(\chi)}$ satisfies the hypothesis of* (i) *for the same $\boldsymbol{v}$ with multiplier λ_χ, then $\mathcal{P}$ admits a schedule supported on the* same *set X, with $c_{g\boldsymbol{v}} = \sum_\chi \lambda_\chi \overline{\chi(g)}$.*

Proof. (i) Write $\sigma = \mu\, A_{\boldsymbol{v}}^{\chi}$ with $\mu \in \mathbb{K}^\times$. By (4.3), $p!\,\sigma = \frac{p!\mu}{|G|} \sum_g \overline{\chi(g)}((g\boldsymbol{v}) \cdot \boldsymbol{z})^p$, which is (1.3) with $\lambda := p!\mu / |G|$. The elements of G producing a given direction $g\boldsymbol{v}$ are the coset $g\,\mathrm{Stab}(\boldsymbol{v})$, so collecting them turns the sum over G into a sum over X with the stated coefficients. Note that these are not all zero: were $\sum_{h\in\mathrm{Stab}(\boldsymbol{v})}\overline{\chi(gh)}=0$ for every g, the same collection would give $A^{\chi}_{\boldsymbol{v}}=0$, contrary to hypothesis. If p is even and $-I\in G$, then $((-\boldsymbol{w})\cdot\boldsymbol{z})^p=(\boldsymbol{w}\cdot\boldsymbol{z})^p$, so antipodal members of X carry identical powers and may be merged, halving the count. (ii) is immediate from (i) because a one-dimensional space contains no directions other than multiples of a fixed nonzero vector, and $\sigma\neq 0$ forces proportionality. (iii) follows by summing the identities of (i) over χ, which changes only the coefficients and not the support. ∎

Theorem 4.3 is the statement that a symmetry group acts as a filter: the $\binom{p+n}{n}$ moment conditions of the design problem collapse to $\dim\mathbb{K}[\boldsymbol{z}]_p^{(\chi)}$ conditions per character, and when that dimension is one there are no conditions at all. The two constructions of this note are the two extremes of the principle.

EXAMPLE 4.4 (The Fourier grid is the abelian case). *Let $G:=U_{m_1}\times\cdots\times U_{m_n}$ act by the diagonal matrices $g_\zeta:=\mathrm{diag}(1,\zeta_1,\dots,\zeta_n)$. These are symmetric, so* (4.1) *reads $(g_\zeta\cdot F)(\boldsymbol{z})=F(g_\zeta\boldsymbol{z})$ and $g_\zeta\cdot\boldsymbol{z}^\beta=\bigl(\prod_{j\ge1}\zeta_j^{\beta_j}\bigr)\boldsymbol{z}^\beta$. Note that G is not orthogonal, which is why Section 4 was set up without that hypothesis. Every monomial spans an isotypic component, with character $\chi_\beta(g_\zeta)=\prod_{j\ge1}\zeta_j^{\beta_j}$, and two monomials of degree p share a component exactly when $m_j\mid\beta_j-\beta'_j$ for all $j\ge1$. When $m_0=\min_j m_j$, the argument around* (2.4) *forces $\beta=\beta'$, so* every *component meeting the symbol is one-dimensional. Take $\boldsymbol{v}:=e_0+e_1+\cdots+e_n$, whose orbit $\{g_\zeta\boldsymbol{v}\}$ is exactly the grid of* (2.2)*. Expanding* (4.3) *and applying* (1.7) *coordinatewise gives $A^{\chi_a}_{\boldsymbol{v}}=\binom{p}{a}\boldsymbol{z}^a$, so Theorem 4.3(ii) reproduces each monomial of the symbol on that one orbit and Theorem 4.3(iii) assembles them, recovering Theorem 2.1. The cost of the abelian choice is visible here: the group has as many one-dimensional components as the grid has points, so the orbit must be as large as the box of monomials it separates, which is $\prod_{j\ge1}m_j$ rather than the far smaller $D_{p,N}$.*

EXAMPLE 4.5 (A nonabelian filter). *Let $G\subset O(3)$ be the icosahedral group of order* 60 *and let $\boldsymbol{v}$ be a vertex of a regular icosahedron inscribed in S^2, whose orbit has* 12 *points forming* 6 *antipodal pairs. The invariant ring of G acting on $\mathbb{R}^3$ is generated in degrees* 2, 6, 10 *and one degree-*15 *element, so its Molien series gives* $\dim\mathbb{R}[\boldsymbol{z}]_4^G=1$*, spanned by $(z\cdot z)^2$, while* $\dim\mathbb{R}[\boldsymbol{z}]_4=15$*. By Theorem 4.3(ii) the single orbit reproduces $(z\cdot z)^2$ with equal weights, so Δ^2 on $\mathbb{R}^3$ has a real schedule of length* 6*. Section 5 shows that* 6 *is minimal.*

5. Powers of the Laplacian are a cubature problem. Example 4.5 is not a coincidence but an instance of a general equivalence. Let $q(\boldsymbol{z}):=\boldsymbol{z}\cdot\boldsymbol{z}$ on $\mathbb{R}^N$, let $\sigma=q^m$ so that $\mathcal{P}=\Delta^m$ and $p=2m$, and let μ denote the uniform probability measure on S^{N-1}.

THEOREM 5.1 (Schedules for Δ^m are spherical cubature rules). *Let $\boldsymbol{v}_1,\dots,\boldsymbol{v}_R\in S^{N-1}$ and $w_1,\dots,w_R\in\mathbb{R}$ with $\sum_r w_r=1$. The following are equivalent.*

(i) *$\sum_{r=1}^R w_r(\boldsymbol{v}_r\cdot\boldsymbol{z})^{2m}=\kappa_{N,m}\,q(\boldsymbol{z})^m$ for all $\boldsymbol{z}\in\mathbb{R}^N$, where $\kappa_{N,m}:=(2m-1)!!\,/\,\bigl(N(N+2)\cdots(N+2m-2)\bigr)$;*

(ii) $\displaystyle\sum_{r=1}^R w_r f(\boldsymbol{v}_r)=\int_{S^{N-1}} f\,d\mu$ *for every $f\in\mathbb{R}[\boldsymbol{v}]_{2m}$;*

(iii) *the rule $(\boldsymbol{v}_r,w_r)$ integrates exactly every polynomial of even degree at most $2m$ on S^{N-1}.*

In particular a real directional schedule for Δ^m with unit directions is the same thing as a cubature rule on S^{N-1} exact in even degrees up to $2m$; if in addition the weights are equal and the node set is centrally symmetric, it is a spherical $(2m{+}1)$-design.

Proof. First we identify the constant. The map $\boldsymbol{z} \mapsto \int_{S^{N-1}} (\boldsymbol{v} \cdot \boldsymbol{z})^{2m} d\mu(\boldsymbol{v})$ is a polynomial of degree $2m$, and it is $O(N)$-invariant because μ is. The ring of $O(N)$-invariant polynomials on $\mathbb{R}^N$ is $\mathbb{R}[q]$, so the integral equals $\kappa\, q^m$ for a constant κ, which we evaluate at a unit vector $\boldsymbol{z} = e$: writing $t := \boldsymbol{v} \cdot e$ for the first coordinate of a uniform point of S^{N-1}, $\kappa = \mathbb{E}[t^{2m}]$. The density of t is proportional to $(1-t^2)^{(N-3)/2}$, and the resulting Beta integral gives $\mathbb{E}[t^{2m}] = (2m-1)!!/\big(N(N+2)\cdots(N+2m-2)\big) = \kappa_{N,m}$.

(i) $\Leftrightarrow$ (ii). For fixed $\boldsymbol{z}$ the function $f_{\boldsymbol{z}}(\boldsymbol{v}) := (\boldsymbol{v}\cdot\boldsymbol{z})^{2m}$ is homogeneous of degree $2m$ in $\boldsymbol{v}$, and $\int f_{\boldsymbol{z}}\, d\mu = \kappa_{N,m} q(\boldsymbol{z})^m$ by the previous paragraph. Hence (i) states precisely that the rule is exact on every $f_{\boldsymbol{z}}$. Both sides of (ii) are linear in f, and $\{f_{\boldsymbol{z}} : \boldsymbol{z} \in \mathbb{R}^N\}$ spans $\mathbb{R}[\boldsymbol{v}]_{2m}$ because powers of linear forms span the space of forms of a given degree. Exactness on a spanning set is exactness on the span, so (i) and (ii) are equivalent.

(ii) $\Leftrightarrow$ (iii). On S^{N-1} we have $q \equiv 1$, so for $k \le m$ and $h \in \mathbb{R}[\boldsymbol{v}]_{2k}$ the function h and the form $q^{m-k}h \in \mathbb{R}[\boldsymbol{v}]_{2m}$ agree at every node and have the same integral; thus exactness in degree $2m$ implies exactness in every even degree at most $2m$, and the converse is trivial.

For the final sentence, Proposition 1.1 converts (i) into a schedule for Δ^m after rescaling the coefficients by $(2m)!/\kappa_{N,m}$, and a centrally symmetric equal-weight rule is exact in odd degrees automatically, so exactness through degree $2m+1$ holds, which is the definition of a spherical $(2m+1)$-design. ◻

THEOREM 5.2 (Lower bound). *$R_{\mathbb{C}}(q^m) \ge \binom{N+m-1}{m} = D_{m,N}$, and hence every directional schedule for Δ^m on $\mathbb{R}^N$, real or complex, has at least $D_{m,N}$ directions.*

Proof. Consider the symmetric bilinear form on $\mathbb{R}[\boldsymbol{y}]_m$ defined by $\langle \Theta, \Psi\rangle := (\Theta\Psi)(\partial)\, q^m$, a scalar because $\Theta\Psi$ has degree $2m = \deg q^m$. Using the integral representation $q^m = \kappa_{N,m}^{-1} \int_{S^{N-1}} (\boldsymbol{v}\cdot\boldsymbol{z})^{2m} d\mu(\boldsymbol{v})$ from the proof of Theorem 5.1 and $(\Theta\Psi)(\partial)(\boldsymbol{v}\cdot\boldsymbol{z})^{2m} = (2m)!\,\Theta(\boldsymbol{v})\Psi(\boldsymbol{v})$,

$$\langle \Theta, \Psi\rangle = \frac{(2m)!}{\kappa_{N,m}} \int_{S^{N-1}} \Theta(\boldsymbol{v})\Psi(\boldsymbol{v})\, d\mu(\boldsymbol{v}),$$

a positive multiple of the $L^2(\mu)$ inner product. It is therefore positive definite on $\mathbb{R}[\boldsymbol{y}]_m$, so the middle catalecticant $\mathrm{Cat}_m(q^m)$, whose associated bilinear form this is, is nonsingular and has rank $D_{m,N}$.

Now let $q^m = \sum_{r=1}^{R} c_r (\boldsymbol{v}_r \cdot \boldsymbol{z})^{2m}$ be any complex decomposition. For $\Theta \in \mathbb{K}[\boldsymbol{y}]_m$, $\Theta(\partial)(\boldsymbol{v}\cdot\boldsymbol{z})^{2m} = \frac{(2m)!}{m!}\Theta(\boldsymbol{v})(\boldsymbol{v}\cdot\boldsymbol{z})^m$, so the image of $\mathrm{Cat}_m(q^m)$ is contained in $\mathrm{span}\{(\boldsymbol{v}_r\cdot \boldsymbol{z})^m : r \le R\}$, a space of dimension at most R. Comparing with the rank just computed gives $R \ge D_{m,N}$. ◻

COROLLARY 5.3 (Tight cases). *For Δ^2 on $\mathbb{R}^3$ the minimum number of directions is 6, attained over $\mathbb{R}$ by the six icosahedral axes with equal positive weights. For Δ^1 on $\mathbb{R}^N$ the minimum is N, attained by any orthonormal frame.*

Proof. Theorem 5.2 gives $R \ge D_{2,3} = 6$ and $R \ge D_{1,N} = N$ respectively. Example 4.5 attains the first, and $q = \sum_j z_j^2$ attains the second. ◻

Three consequences are worth stating plainly. The equivalence removes the search: for Δ^m one does not look for Waring decompositions but reads off a cubature rule, of which large tabulated families exist in every dimension. The nodes are real and the weights are positive, so no complex arithmetic is needed for polyharmonic operators, in contrast to the roots-of-unity schedule. And the classical lower bound for centrally symmetric cubature coincides with the catalecticant bound of Theorem 5.2, so the two literatures agree on what is possible.

Remark 5.4 (The bound is not always attained over $\mathbb{R}$). *Attainment is a genuinely finer question than the bound. A numerical search over real nodes, with the weights eliminated by orthogonal projection so that only the nodes are optimized, reaches $D_{m,N}$ for $(N,m) = (3,1)$, $(3,2)$ and $(4,1)$. For $(N,m) = (3,3)$ and $(4,2)$ it instead stalls at $D_{m,N} + 1$, namely 11 nodes against the bound 10 in both cases, with positive weights. The evidence that the bound is genuinely missed rather than merely unfound is that at $R = D_{m,N}$ the residual settles near 10^{-1} over forty restarts, whereas at $R = D_{m,N} + 1$ it reaches machine precision on the first attempt. This is consistent with the general fact that real Waring rank can exceed complex Waring rank, and with Theorem 4.3: in degree 6 the icosahedral group has a two-dimensional invariant space, so a single orbit no longer suffices. It is not a proof, since the search is nonconvex.*

6. One direction set for many operators. The implementation described in the main paper schedules an operator by expanding it into monomials and scheduling each monomial separately. The following elementary observations quantify what that costs and identify the right object to optimize.

Definition 6.1 (Shared schedule). *Let $\mathcal{P}_1, \dots, \mathcal{P}_K$ have symbols $\sigma_1, \dots, \sigma_K \in \mathbb{K}[\boldsymbol{z}]_p$ and put $U := \operatorname{span}_{\mathbb{K}}\{\sigma_1, \dots, \sigma_K\}$. A* shared schedule *is a finite set $X = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_R\}$ together with coefficient vectors $c^{(i)} \in \mathbb{K}^R$ such that $(c_r^{(i)}, \boldsymbol{v}_r)_r$ is a schedule for $\mathcal{P}_i$, for each i. Write $R_{\mathbb{K}}(U)$ for the least such R.*

Proposition 6.2 (Shared rank). *With $W_X := \operatorname{span}_{\mathbb{K}}\{(\boldsymbol{v} \cdot \boldsymbol{z})^p : \boldsymbol{v} \in X\}$, a shared schedule on X exists if and only if $U \subseteq W_X$. Moreover*

$$\max\Big(\dim_{\mathbb{K}} U,\ \max_i R_{\mathbb{K}}(\sigma_i)\Big) \ \le\ R_{\mathbb{K}}(U) \ \le\ \min\Big(\sum_i R_{\mathbb{K}}(\sigma_i),\ D_{p,N}\Big). \tag{6.1}$$

Proof. By Proposition 1.1 a schedule for $\mathcal{P}_i$ on X exists exactly when $\sigma_i \in W_X$, and this for all i is $U \subseteq W_X$ since W_X is a subspace. For the lower bounds, $\dim W_X \le |X|$ gives $\dim U \le R_{\mathbb{K}}(U)$, and $U \subseteq W_X$ contains each σ_i so $|X| \ge R_{\mathbb{K}}(\sigma_i)$. For the upper bounds, the union of individually minimal sets is admissible, and any X of size $D_{p,N}$ whose power images are independent has $W_X = \mathbb{K}[\boldsymbol{z}]_p \supseteq U$. ∎

Corollary 6.3 (The full derivative tensor). *Taking $U = \mathbb{K}[\boldsymbol{z}]_p$, that is, requiring every partial derivative of order p in the N active variables, gives*

$$R_{\mathbb{K}}(\mathbb{K}[\boldsymbol{z}]_p) \ =\ D_{p,N} \ =\ \binom{p+N-1}{N-1}, \tag{6.2}$$

attained by any X of that cardinality whose power images are linearly independent, hence by a generic X. No optimization is involved and no smaller set can work.

Proof. Both bounds of (6.1) equal $D_{p,N}$. Independence of $\{(\boldsymbol{v}_r \cdot \boldsymbol{z})^p\}$ for generic $\boldsymbol{v}_r$ holds because the Veronese variety is nondegenerate, so no nonzero linear functional on $\mathbb{K}[\boldsymbol{z}]_p$ annihilates all pth powers. ∎

Corollary 6.3 is the sharpest available answer to the efficiency question when several derivatives are wanted at once: the correct cost is the dimension of what is being asked for, not the sum of the individual ranks. The gap between the two is large already in modest cases; Table 7.1 records it.

Remark 6.4 (Conditioning is a separate objective). *Minimal length does not imply minimal cost. Let M_X be the matrix of the map $c \mapsto \sum_r c_r(\boldsymbol{v}_r \cdot \boldsymbol{z})^p$ in the monomial basis; the coefficients of a schedule are obtained by solving with M_X, so the*

Table 7.1

Directions required, by construction. The grid is Theorem 2.1 in the given coordinates; term-by-term sums the monomial ranks of the main paper; the bound is Theorem 5.2 for Δ^m and Corollary 6.3 for the full tensor; the last column is the shortest real schedule found by numerical search over nodes, with weights eliminated by orthogonal projection.

target	N	grid	term-by-term	bound	search	weights
Δ^1	3	9	3	3	3	positive
Δ^2	3	25	12	6	6	positive, equal
Δ^3	3	49	42	10	11	positive
Δ^1	4	27	4	4	4	positive
Δ^2	4	125	22	10	11	positive
all order 4	3	—	54	15	15	—
all order 6	3	—	192	28	28	—
all order 4	4	—	150	35	35	—
all order 6	4	—	776	84	84	—

relative error of the assembled derivative is governed by the condition number of M_X. The Fourier grid of Theorem 2.1 is a discrete Fourier matrix up to diagonal scaling and is extremely well conditioned, whereas generic minimal sets are not; Table 7.3 measures both. Orbit constructions are attractive precisely because they achieve short length and equal weights simultaneously, the icosahedral schedule of Corollary 5.3 being minimal and perfectly balanced at once. A realistic objective is therefore a weighted count of real and complex jet passes subject to a conditioning constraint, rather than the raw Waring rank.

7. Numerical corroboration. Every statement above that asserts a specific number has been checked numerically by the script `verify_theory_directions.py` in `experiments/tools`, which also reproduces the tables of this section. Table 7.1 compares, for Δ^m and for the full derivative tensor, the length of the Fourier grid of Theorem 2.1, the term-by-term total used by the current implementation, the lower bound of Theorem 5.2 or Corollary 6.3, and the shortest real schedule located by direct search.

Three readings of Table 7.1 matter. First, the universal grid is a poor choice for a structured operator: at Δ^2 on $\mathbb{R}^3$ it needs 25 directions where the term-by-term expansion needs 12, so Theorem 2.1 is a fallback for symbols about which nothing is known, not a replacement for the monomial schedule. Second, term-by-term is nonetheless not optimal once the operator couples coordinates: the same Δ^2 needs only the six real icosahedral axes, which Corollary 5.3 certifies as minimal, and Δ^3 on $\mathbb{R}^3$ drops from 42 to 11. The saving grows with m because the term-by-term total grows like the number of monomials times their individual ranks while the bound grows only like $D_{m,N}$. Third, the largest gap appears when many derivatives are wanted at once: the full order-six tensor in three variables costs 192 directions term by term against the exact optimum 28 of Corollary 6.3, which requires no search at all.

Table 7.2 records the rules that the accompanying implementation actually builds for Δ^m, following Theorem 5.1. They come from orbits of the group of signed permutations, so by Theorem 4.3 the weights follow from one linear condition per invariant of degree $2m$ rather than from a search, and the resulting nodes are real with weights

TABLE 7.2

Rules built by the implementation for Δ^m on $\mathbb{R}^d$, against the lower bound of Theorem 5.2 and the term-by-term total. Entries marked $\star$ meet the bound.

	$m=1$			$m=2$			$m=3$			$m=4$	
d	rule	bnd	t-b-t	rule	bnd	t-b-t	rule	bnd	t-b-t	rule	bnd
3	$3^\star$	3	3	$6^\star$	6	12	13	10	42	25	15
4	$4^\star$	4	4	12	10	22	24	20	100	64	35
5	$5^\star$	5	5	21	15	35	41	35	195	121	70
6	$6^\star$	6	6	36	21	51	68	56	336	208	126

TABLE 7.3

Length against conditioning for a symbol of degree p in N variables: a generic minimal set attaining Corollary 6.3 versus the Fourier grid of Theorem 2.1.

p	N	generic minimal	$\kappa(M_X)$	grid	$\kappa(M_X)$
4	2	5	2.0×10^2	5	6.0×10^0
6	2	7	1.2×10^2	7	2.0×10^1
4	3	15	4.9×10^2	25	1.2×10^1
6	3	28	2.2×10^4	49	9.0×10^1

that are positive in all but four of the sixteen tabulated cases. Two families of entries meet the bound of Theorem 5.2 exactly: an orthonormal frame at $m = 1$ in every dimension, and the icosahedral axes at $(d, m) = (3, 2)$. Elsewhere the constructive rule sits above the shortest schedule that numerical search can find, by two directions at $(3, 3)$ and one at $(4, 2)$, which is the price of insisting on a deterministic construction with a linear solve. Beyond the orders tabulated here the orbits considered supply too few invariant conditions, and the implementation raises rather than returning an approximation.

Table 7.3 quantifies Remark 6.4. The grid buys its extra length back in stability, which is the reason it should not be discarded even where shorter schedules exist.

8. Summary. The results assemble into a selection rule for building a schedule, ordered by how much structure the symbol has. If nothing is known about σ beyond its coordinate degrees, Theorem 2.1 applies unconditionally, needs no solve, and is well conditioned, at length $\prod_{j\geq 1} m_j$. If σ is projectively equivalent to a monomial, running the same construction in the coordinates that make it a monomial is minimal, by Remark 3.5; this covers the polyharmonic operator in two variables. If σ is a power of a nondegenerate quadratic form, Theorem 5.1 replaces the design problem by spherical cubature, giving real nodes, positive weights and, in the smallest cases, provable minimality. If σ is invariant under any finite group, Theorem 4.3 reduces the moment conditions to the dimension of the relevant isotypic component. And if several operators or a whole derivative tensor are wanted, Corollary 6.3 gives an exactly optimal shared set of length equal to the dimension of the span, with no search.

Of these, the cubature branch is implemented: `apolarity.cubature` builds the rules of Table 7.2 and routes Δ^m in three or more variables through them, while two variables continue to use the closed form.

What is not settled is the general case. For a symbol with no symmetry and no quadratic structure in $N \geq 3$ active variables, no constructive procedure is known that attains $R_{\mathbb{C}}(\sigma)$; computing the rank is already NP-hard, and numerical decomposition can fail because the border rank may be strictly smaller. Corollary 3.4 makes the stakes concrete: the universal grid is off by a factor tending to $N!$, so the gap is worth closing wherever structure permits, and the four structured cases above are where it does.